Alignment Verification Booklet

Revised Stimulus Selection - Phase 2C

MSc dissertation project

Project: intent2control-dissertation

Generation date: 2026-08-03

Methodology status: Awaiting manual/supervisor alignment review

Total triplets: 8

Total recommended mixes: 24

This booklet consolidates Phase 2C quality-assurance evidence only. The selection recommendations have not been changed, and no triplet is finally approved unless the manual review table records approval.

Executive Summary

- 8 triplets verified.
- 24 pairwise comparisons.
- 8 rapid-switch files.
- 16 visual figures.
- Maximum automatic offset: 58.050 ms.
- Maximum-offset triplet: Pouring Room / Wide Ratings.
- Automatic classifications are screening results. Final inclusion requires visual inspection of waveform figures and perceptual inspection of rapid-switch audio.
- REVIEW and FAIL labels mean manual inspection is required, not necessarily that the mixes must be discarded.

Status	Triplets
PASS	Red To Blue / Wide Ratings
REVIEW	In The Meantime / Wide Ratings Pouring Room / Similar Ratings Red To Blue / Similar Ratings
FAIL	In The Meantime / Similar Ratings Lead Me / Similar Ratings Lead Me / Wide Ratings Pouring Room / Wide Ratings

Methods Summary

- Approved 28-second excerpts were retained.
- Pairwise residual alignment was recomputed for each recommended triplet.
- Lag was reported in samples and milliseconds.
- Waveform plots were inspected on a shared time axis.
- Zoomed transient plots were generated.
- Rapid-switch WAVs were created for direct perceptual comparison.
- No loudness normalisation was applied to review audio.
- No selection decisions were changed.
- Automatic PASS/REVIEW/FAIL thresholds: PASS <=10 ms and correlation >=0.60; REVIEW <=30 ms and correlation >=0.35; otherwise FAIL.
- Automatic confidence is not treated as sufficient on its own.

Overall Alignment Results

Song	Rating condition	Original mix names	Max abs lag ms	Min corr.	Auto status	Rapid-switch	Waveform	Zoomed	Manual	Notes
In The Meantime	Similar Ratings	QUT-B | DU-H | DU-I	34.830	0.555	FAIL	review_audio/In The Meantime/Similar Ratings/RapidSwitch.wav	figures/In The Meantime/Similar Ratings/stacked_waveforms.png	figures/In The Meantime/Similar Ratings/strongest_transient_zoom.png	pending	requires_alignment_review_before_supervisor_review
In The Meantime	Wide Ratings	DU-K | QUT-pro | DU-N	23.220	0.682	REVIEW	review_audio/In The Meantime/Wide Ratings/RapidSwitch.wav	figures/In The Meantime/Wide Ratings/stacked_waveforms.png	figures/In The Meantime/Wide Ratings/strongest_transient_zoom.png	pending	suitable_for_supervisor_review
Lead Me	Similar Ratings	DU-D | DU-E | QUT-F	46.440	0.460	FAIL	review_audio/Lead Me/Similar Ratings/RapidSwitch.wav	figures/Lead Me/Similar Ratings/stacked_waveforms.png	figures/Lead Me/Similar Ratings/strongest_transient_zoom.png	pending	requires_alignment_review_before_supervisor_review
Lead Me	Wide Ratings	PXL-L1 | PXL-L4 | McG-pro2	46.440	0.492	FAIL	review_audio/Lead Me/Wide Ratings/RapidSwitch.wav	figures/Lead Me/Wide Ratings/stacked_waveforms.png	figures/Lead Me/Wide Ratings/strongest_transient_zoom.png	pending	requires_alignment_review_before_supervisor_review
Pouring Room	Similar Ratings	McG-R | McG-T | McG-X	0.000	0.364	REVIEW	review_audio/Pouring Room/Similar Ratings/RapidSwitch.wav	figures/Pouring Room/Similar Ratings/stacked_waveforms.png	figures/Pouring Room/Similar Ratings/strongest_transient_zoom.png	pending	suitable_for_supervisor_review
Pouring Room	Wide Ratings	McG-pro1 | McG-U | McG-V	58.050	0.286	FAIL	review_audio/Pouring Room/Wide Ratings/RapidSwitch.wav	figures/Pouring Room/Wide Ratings/stacked_waveforms.png	figures/Pouring Room/Wide Ratings/strongest_transient_zoom.png	pending	requires_alignment_review_before_supervisor_review
Red To Blue	Similar Ratings	McG-B | McG-G | McG-F	11.610	0.575	REVIEW	review_audio/Red To Blue/Similar Ratings/RapidSwitch.wav	figures/Red To Blue/Similar Ratings/stacked_waveforms.png	figures/Red To Blue/Similar Ratings/strongest_transient_zoom.png	pending	suitable_for_supervisor_review
Red To Blue	Wide Ratings	McG-C | McG-H | McG-pro1	0.000	0.710	PASS	review_audio/Red To Blue/Wide Ratings/RapidSwitch.wav	figures/Red To Blue/Wide Ratings/stacked_waveforms.png	figures/Red To Blue/Wide Ratings/strongest_transient_zoom.png	pending	suitable_for_supervisor_review

In The Meantime - Similar Ratings

Original mix names: QUT-B | DU-H | DU-I | Automatic status: FAIL

Maximum lag: 34.830 ms
Minimum correlation: 0.555
Automatic confidence/result: 0.555 / FAIL
Stereo-imbalance QC flags: QUT-B: false; DU-H: false; DU-I: false
Rapid-switch audio path: review_audio/In The Meantime/Similar Ratings/RapidSwitch.wav

Pair	Lag samples	Lag ms	Correlation	Automatic result
QUT-B-DU-H	0	0.000	0.582	REVIEW
QUT-B-DU-I	-1024	-23.220	0.555	REVIEW
DU-H-DU-I	-1536	-34.830	0.617	FAIL

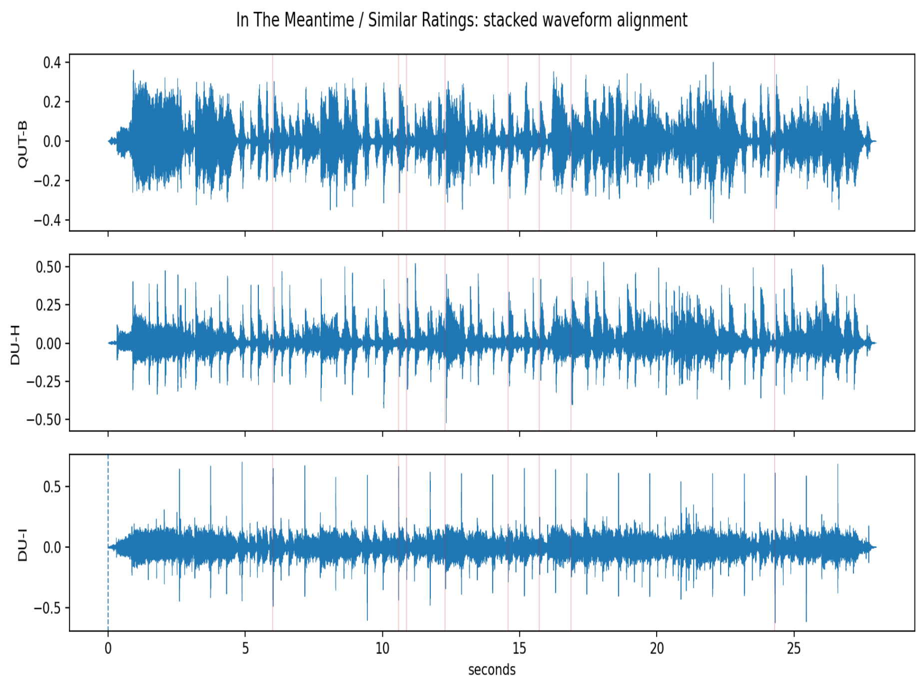
Review-audio paths relative to 05_alignment_verification
review_audio/In The Meantime/Similar Ratings/QUT-B_28sec.wav
review_audio/In The Meantime/Similar Ratings/DU-H_28sec.wav
review_audio/In The Meantime/Similar Ratings/DU-I_28sec.wav
review_audio/In The Meantime/Similar Ratings/RapidSwitch.wav

Manual-review prompts

- Do major transients align visually?
- Is there any apparent constant offset?
- Is there any timing drift across the 28-second excerpt?
- Does rapid switching reveal a perceptible timing jump?
- Is the issue alignment, or only a mix-production difference?
- Accept alignment?
- Reviewer comments.

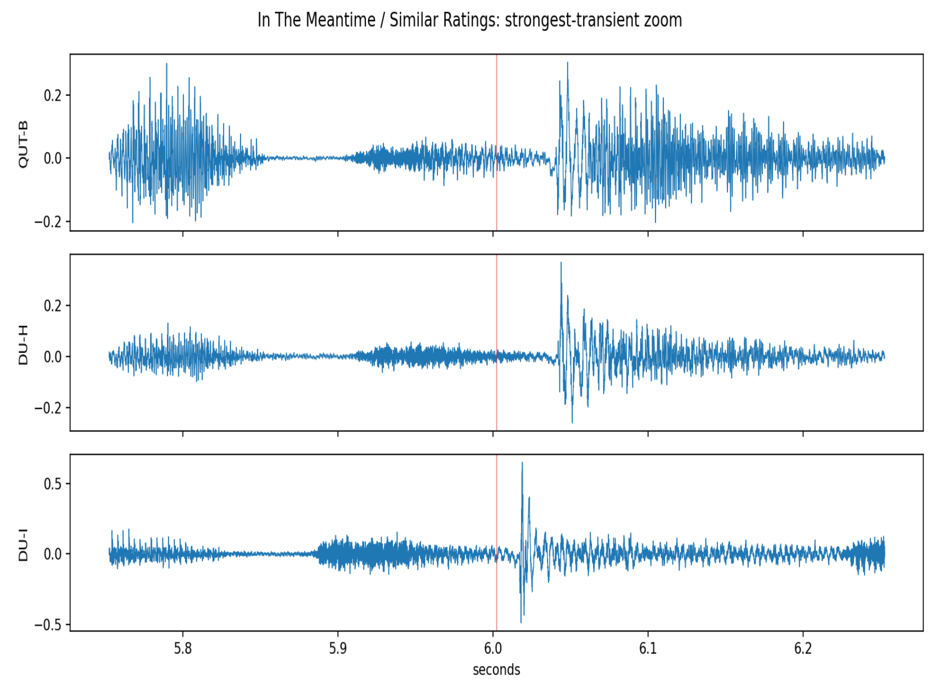
Visual Evidence: In The Meantime - Similar Ratings

Full stacked-waveform figure



figures/In The Meantime/Similar Ratings/stacked_waveforms.png

Zoomed strongest-transient figure



figures/In The Meantime/Similar Ratings/strongest_transient_zoom.png

In The Meantime - Wide Ratings

Original mix names: DU-K | QUT-pro | DU-N | Automatic status: REVIEW

Maximum lag: 23.220 ms
Minimum correlation: 0.682
Automatic confidence/result: 0.682 / REVIEW
Stereo-imbalance QC flags: DU-K: false; QUT-pro: false; DU-N: false
Rapid-switch audio path: review_audio/In The Meantime/Wide Ratings/RapidSwitch.wav

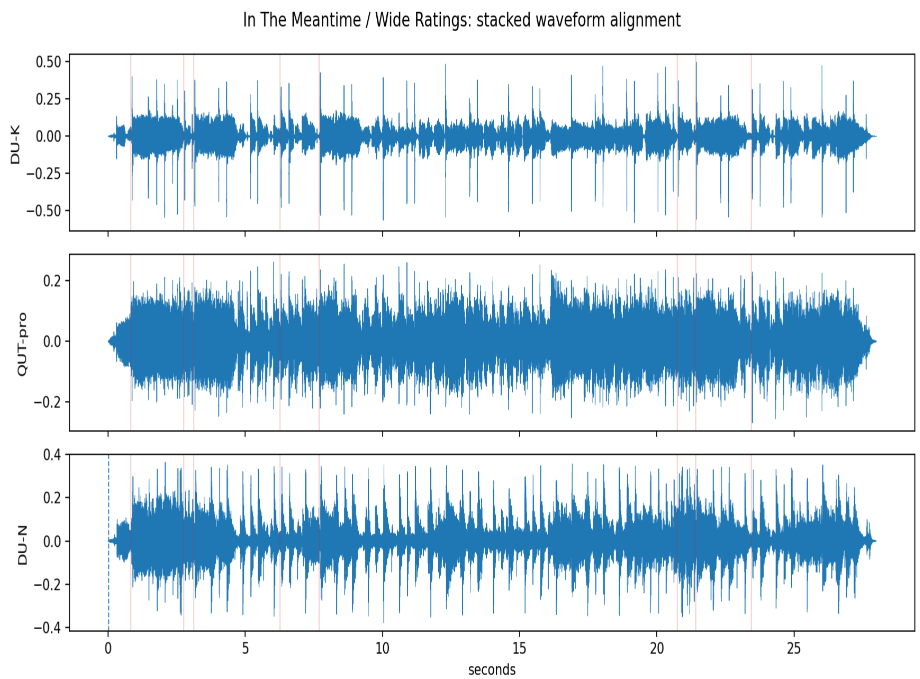
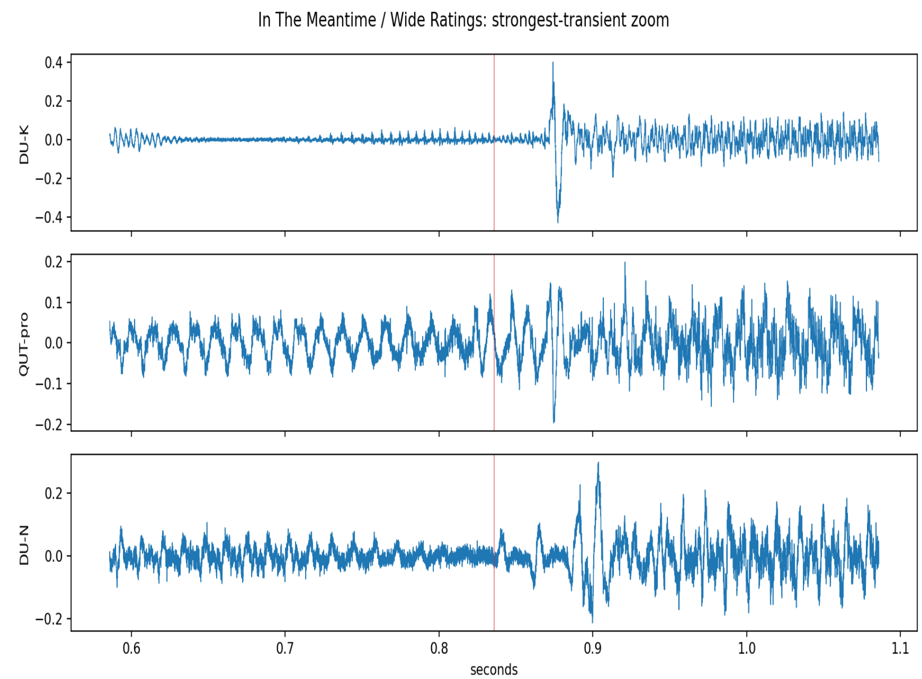
Pair	Lag samples	Lag ms	Correlation	Automatic result
DU-K-QUT-pro	0	0.000	0.704	PASS
DU-K-DU-N	1024	23.220	0.682	REVIEW
QUT-pro-DU-N	1024	23.220	0.719	REVIEW

Review-audio paths relative to 05_alignment_verification
review_audio/In The Meantime/Wide Ratings/DU-K_28sec.wav
review_audio/In The Meantime/Wide Ratings/QUT-pro_28sec.wav
review_audio/In The Meantime/Wide Ratings/DU-N_28sec.wav
review_audio/In The Meantime/Wide Ratings/RapidSwitch.wav

Manual-review prompts

- Do major transients align visually?
- Is there any apparent constant offset?
- Is there any timing drift across the 28-second excerpt?
- Does rapid switching reveal a perceptible timing jump?
- Is the issue alignment, or only a mix-production difference?
- Accept alignment?
- Reviewer comments.

Visual Evidence: In The Meantime - Wide Ratings

Full stacked-waveform figure	Zoomed strongest-transient figure
	
figures/In The Meantime/Wide Ratings/stacked_waveforms.png	figures/In The Meantime/Wide Ratings/strongest_transient_zoom.png

Lead Me - Similar Ratings

Original mix names: DU-D | DU-E | QUT-F | Automatic status: FAIL

Maximum lag: 46.440 ms
Minimum correlation: 0.460
Automatic confidence/result: 0.460 / FAIL
Stereo-imbalance QC flags: DU-D: false; DU-E: false; QUT-F: false
Rapid-switch audio path: review_audio/Lead Me/Similar Ratings/RapidSwitch.wav

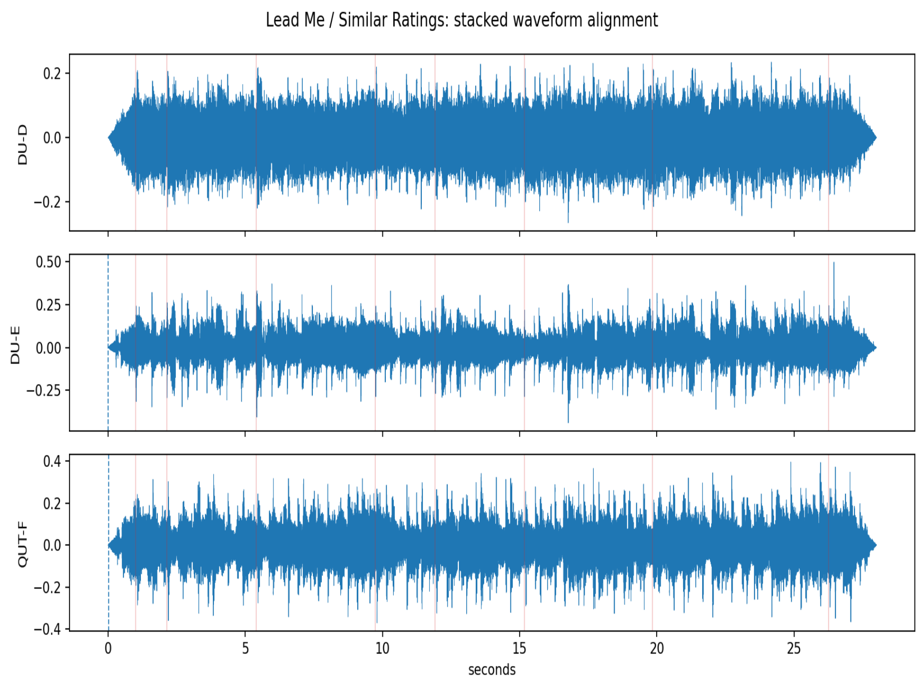
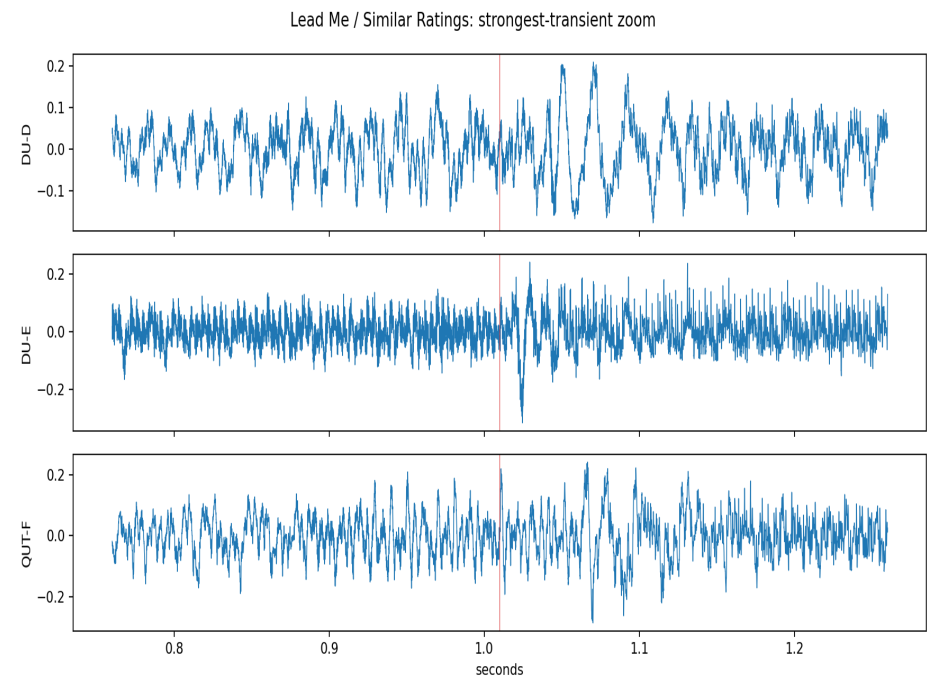
Pair	Lag samples	Lag ms	Correlation	Automatic result
DU-D-DU-E	-1024	-23.220	0.460	REVIEW
DU-D-QUT-F	1024	23.220	0.543	REVIEW
DU-E-QUT-F	2048	46.440	0.599	FAIL

Review-audio paths relative to 05_alignment_verification
review_audio/Lead Me/Similar Ratings/DU-D_28sec.wav
review_audio/Lead Me/Similar Ratings/DU-E_28sec.wav
review_audio/Lead Me/Similar Ratings/QUT-F_28sec.wav
review_audio/Lead Me/Similar Ratings/RapidSwitch.wav

Manual-review prompts

- Do major transients align visually?
- Is there any apparent constant offset?
- Is there any timing drift across the 28-second excerpt?
- Does rapid switching reveal a perceptible timing jump?
- Is the issue alignment, or only a mix-production difference?
- Accept alignment?
- Reviewer comments.

Visual Evidence: Lead Me - Similar Ratings

Full stacked-waveform figure	Zoomed strongest-transient figure
	
figures/Lead Me/Similar Ratings/stacked_waveforms.png	figures/Lead Me/Similar Ratings/strongest_transient_zoom.png

Lead Me - Wide Ratings

Original mix names: PXL-L1 | PXL-L4 | McG-pro2 | Automatic status: FAIL

Maximum lag: 46.440 ms
Minimum correlation: 0.492
Automatic confidence/result: 0.492 / FAIL
Stereo-imbalance QC flags: PXL-L1: false; PXL-L4: false; McG-pro2: false
Rapid-switch audio path: review_audio/Lead Me/Wide Ratings/RapidSwitch.wav

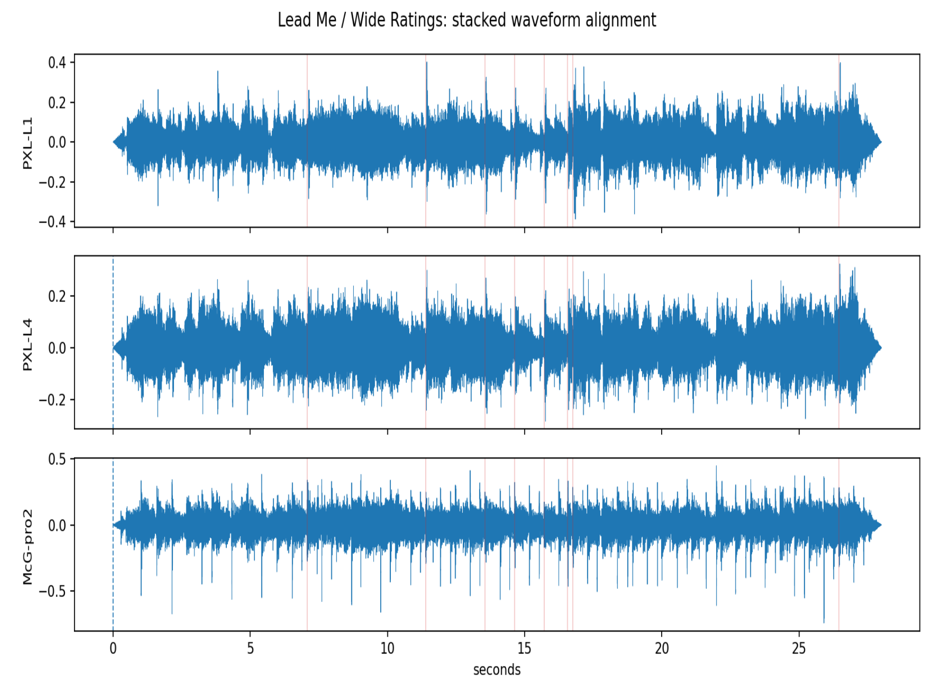
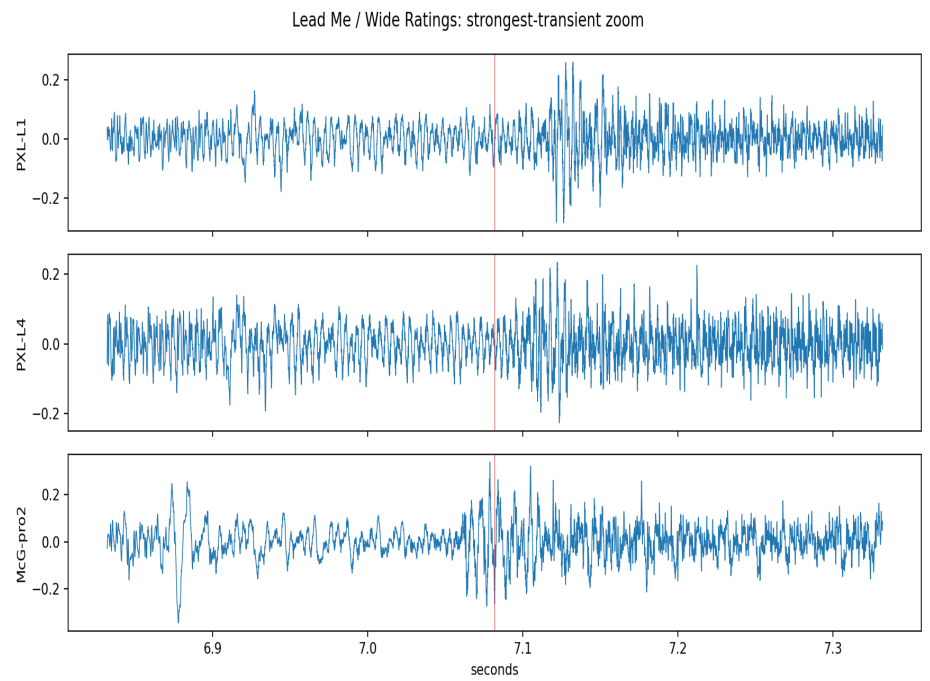
Pair	Lag samples	Lag ms	Correlation	Automatic result
PXL-L1-PXL-L4	-512	-11.610	0.833	REVIEW
PXL-L1-McG-pro2	-2048	-46.440	0.492	FAIL
PXL-L4-McG-pro2	-1536	-34.830	0.600	FAIL

Review-audio paths relative to 05_alignment_verification
review_audio/Lead Me/Wide Ratings/PXL-L1_28sec.wav
review_audio/Lead Me/Wide Ratings/PXL-L4_28sec.wav
review_audio/Lead Me/Wide Ratings/McG-pro2_28sec.wav
review_audio/Lead Me/Wide Ratings/RapidSwitch.wav

Manual-review prompts

- Do major transients align visually?
- Is there any apparent constant offset?
- Is there any timing drift across the 28-second excerpt?
- Does rapid switching reveal a perceptible timing jump?
- Is the issue alignment, or only a mix-production difference?
- Accept alignment?
- Reviewer comments.

Visual Evidence: Lead Me - Wide Ratings

Full stacked-waveform figure	Zoomed strongest-transient figure
	
figures/Lead Me/Wide Ratings/stacked_waveforms.png	figures/Lead Me/Wide Ratings/strongest_transient_zoom.png

Pouring Room - Similar Ratings

Original mix names: McG-R | McG-T | McG-X | Automatic status: REVIEW

Maximum lag: 0.000 ms
Minimum correlation: 0.364
Automatic confidence/result: 0.364 / REVIEW
Stereo-imbalance QC flags: McG-R: false; McG-T: false; McG-X: false
Rapid-switch audio path: review_audio/Pouring Room/Similar Ratings/RapidSwitch.wav

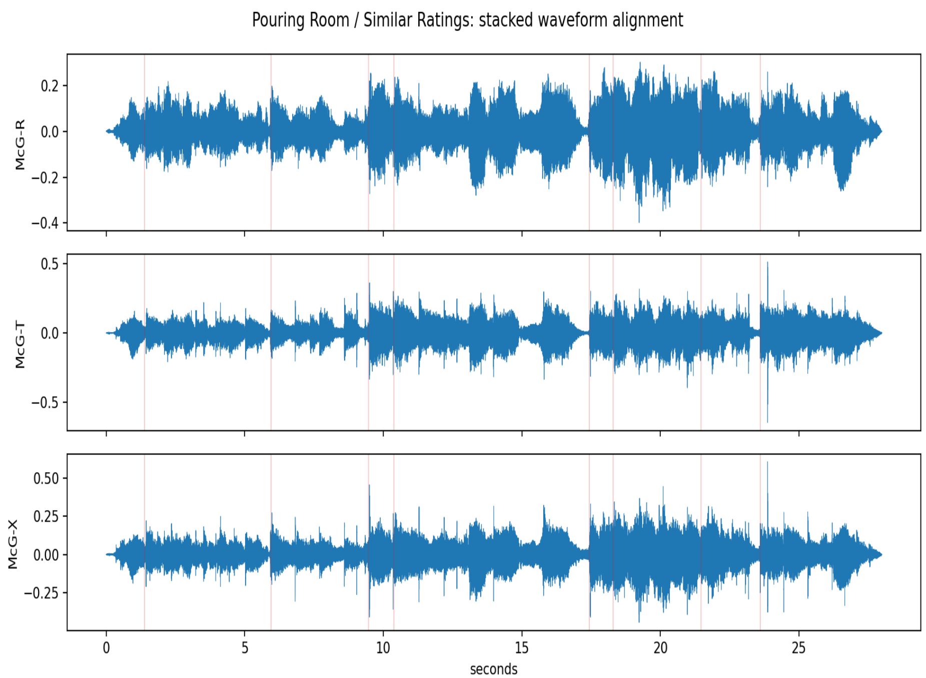
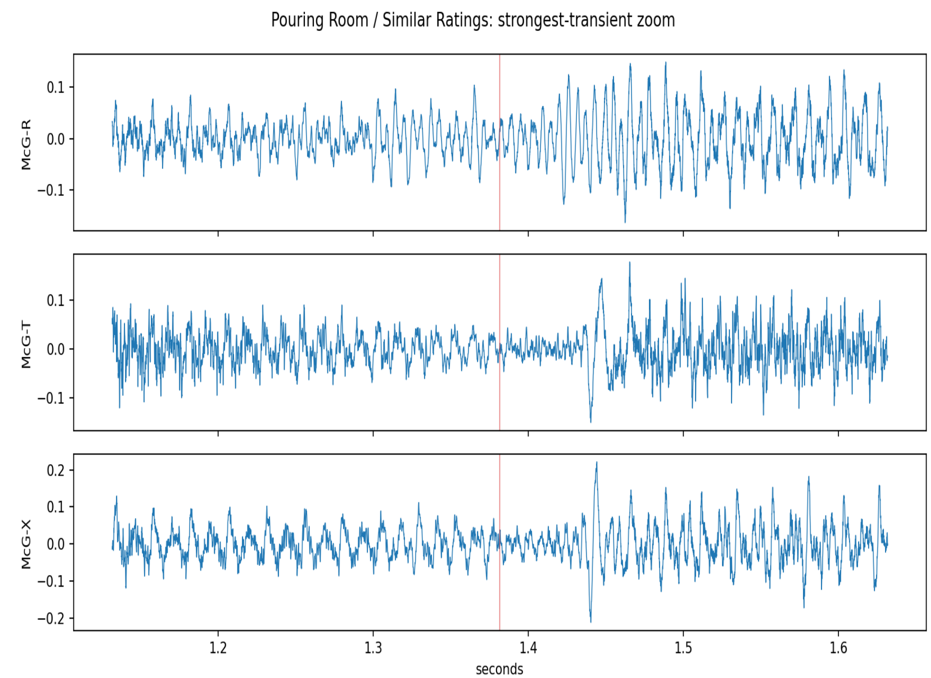
Pair	Lag samples	Lag ms	Correlation	Automatic result
McG-R-McG-T	0	0.000	0.364	REVIEW
McG-R-McG-X	0	0.000	0.389	REVIEW
McG-T-McG-X	0	0.000	0.765	PASS

Review-audio paths relative to 05_alignment_verification
review_audio/Pouring Room/Similar Ratings/McG-R_28sec.wav
review_audio/Pouring Room/Similar Ratings/McG-T_28sec.wav
review_audio/Pouring Room/Similar Ratings/McG-X_28sec.wav
review_audio/Pouring Room/Similar Ratings/RapidSwitch.wav

Manual-review prompts

- Do major transients align visually?
- Is there any apparent constant offset?
- Is there any timing drift across the 28-second excerpt?
- Does rapid switching reveal a perceptible timing jump?
- Is the issue alignment, or only a mix-production difference?
- Accept alignment?
- Reviewer comments.

Visual Evidence: Pouring Room - Similar Ratings

<table>
<tr><th>Full stacked-waveform figure</th><th>Zoomed strongest-transient figure</th></tr>
<tr><td>
</td><td>
</td></tr>
<tr><td>figures/Pouring Room/Similar Ratings/stacked_waveforms.png</td><td>figures/Pouring Room/Similar Ratings/strongest_transient_zoom.png</td></tr>
</table>

Pouring Room - Wide Ratings

Original mix names: McG-pro1 | McG-U | McG-V | Automatic status: FAIL

Maximum lag: 58.050 ms
Minimum correlation: 0.286
Automatic confidence/result: 0.286 / FAIL
Stereo-imbalance QC flags: McG-pro1: false; McG-U: false; McG-V: false
Rapid-switch audio path: review_audio/Pouring Room/Wide Ratings/RapidSwitch.wav

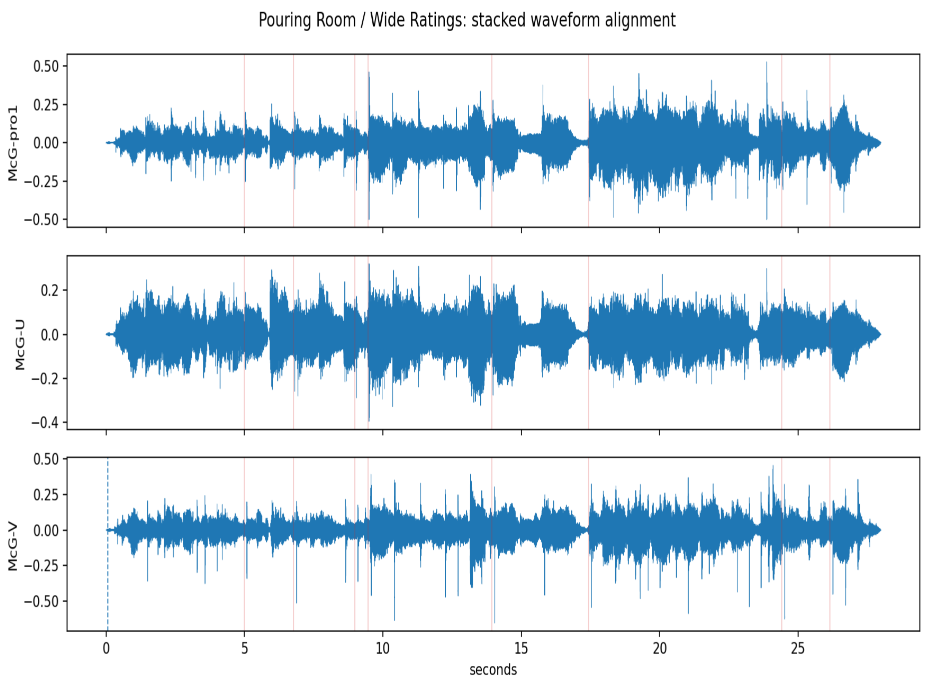
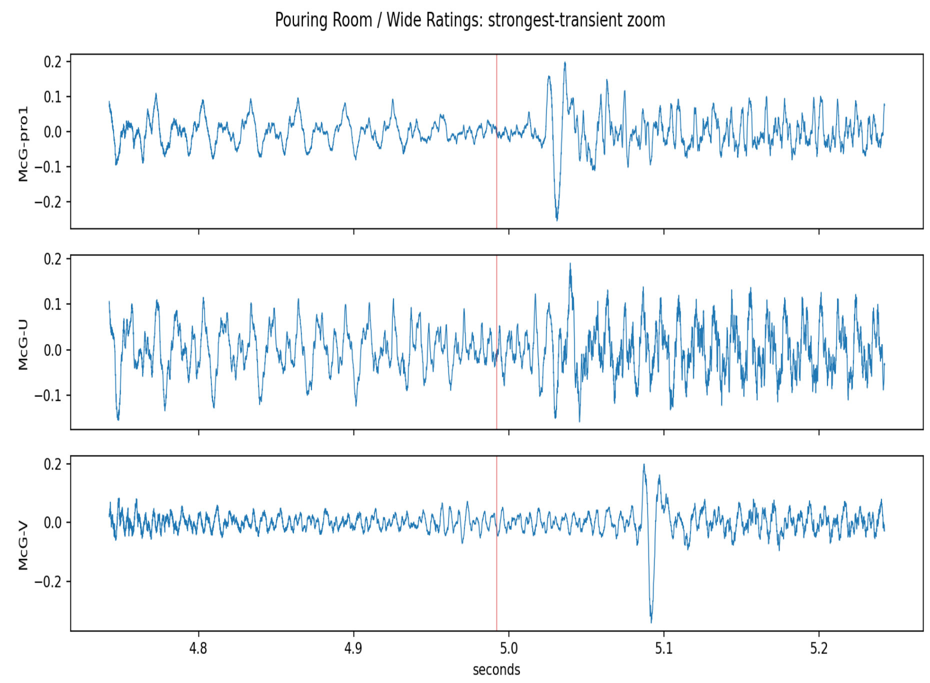
Pair	Lag samples	Lag ms	Correlation	Automatic result
McG-pro1-McG-U	0	0.000	0.494	REVIEW
McG-pro1-McG-V	2560	58.050	0.670	FAIL
McG-U-McG-V	2560	58.050	0.286	FAIL

Review-audio paths relative to 05_alignment_verification
review_audio/Pouring Room/Wide Ratings/McG-pro1_28sec.wav
review_audio/Pouring Room/Wide Ratings/McG-U_28sec.wav
review_audio/Pouring Room/Wide Ratings/McG-V_28sec.wav
review_audio/Pouring Room/Wide Ratings/RapidSwitch.wav

Manual-review prompts

- Do major transients align visually?
- Is there any apparent constant offset?
- Is there any timing drift across the 28-second excerpt?
- Does rapid switching reveal a perceptible timing jump?
- Is the issue alignment, or only a mix-production difference?
- Accept alignment?
- Reviewer comments.

Visual Evidence: Pouring Room - Wide Ratings

Full stacked-waveform figure	Zoomed strongest-transient figure
	
figures/Pouring Room/Wide Ratings/stacked_waveforms.png	figures/Pouring Room/Wide Ratings/strongest_transient_zoom.png

Red To Blue - Similar Ratings

Original mix names: McG-B | McG-G | McG-F | Automatic status: REVIEW

Maximum lag: 11.610 ms
Minimum correlation: 0.575
Automatic confidence/result: 0.575 / REVIEW
Stereo-imbalance QC flags: McG-B: true; McG-G: false; McG-F: false
Rapid-switch audio path: review_audio/Red To Blue/Similar Ratings/RapidSwitch.wav

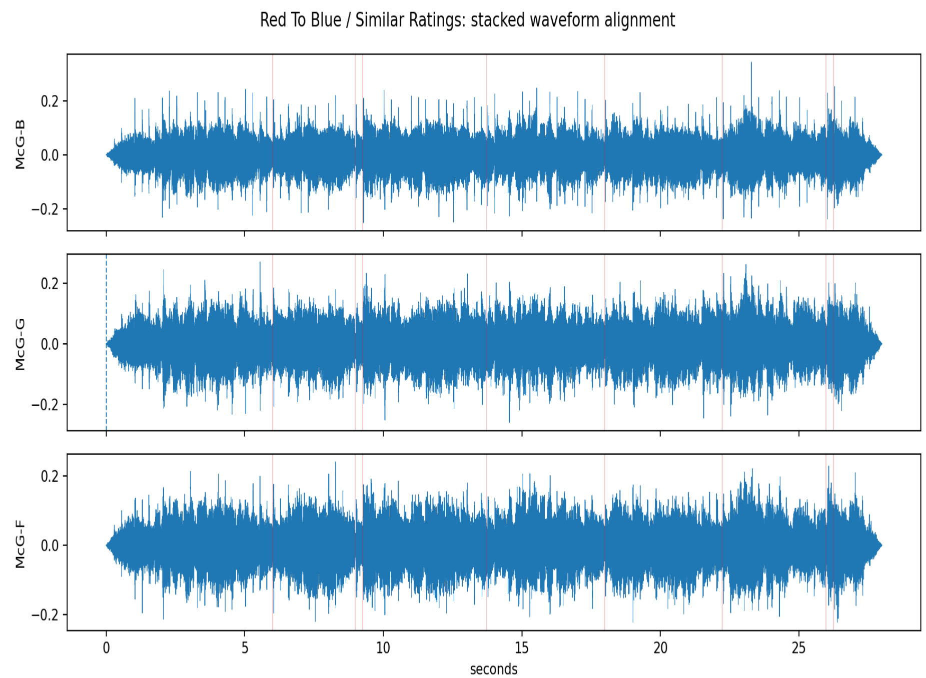
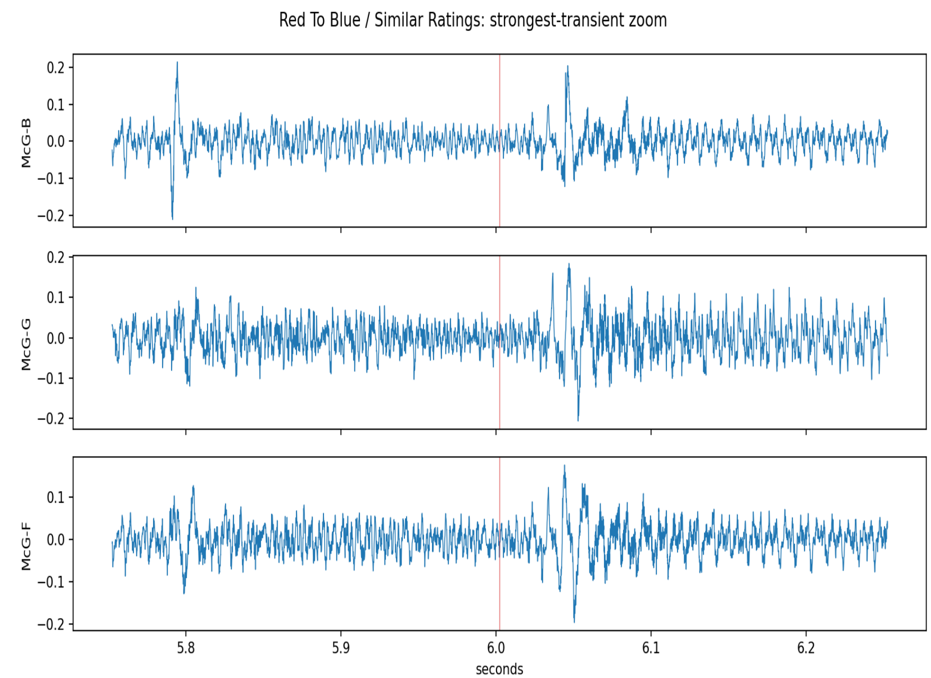
Pair	Lag samples	Lag ms	Correlation	Automatic result
McG-B-McG-G	512	11.610	0.575	REVIEW
McG-B-McG-F	0	0.000	0.685	PASS
McG-G-McG-F	0	0.000	0.621	PASS

Review-audio paths relative to 05_alignment_verification
review_audio/Red To Blue/Similar Ratings/McG-B_28sec.wav
review_audio/Red To Blue/Similar Ratings/McG-G_28sec.wav
review_audio/Red To Blue/Similar Ratings/McG-F_28sec.wav
review_audio/Red To Blue/Similar Ratings/RapidSwitch.wav

Manual-review prompts

- Do major transients align visually?
- Is there any apparent constant offset?
- Is there any timing drift across the 28-second excerpt?
- Does rapid switching reveal a perceptible timing jump?
- Is the issue alignment, or only a mix-production difference?
- Accept alignment?
- Reviewer comments.

Visual Evidence: Red To Blue - Similar Ratings

Full stacked-waveform figure	Zoomed strongest-transient figure
	
figures/Red To Blue/Similar Ratings/stacked_waveforms.png	figures/Red To Blue/Similar Ratings/strongest_transient_zoom.png

Red To Blue - Wide Ratings

Original mix names: McG-C | McG-H | McG-pro1 | Automatic status: PASS

Maximum lag: 0.000 ms
Minimum correlation: 0.710
Automatic confidence/result: 0.710 / PASS
Stereo-imbalance QC flags: McG-C: false; McG-H: false; McG-pro1: false
Rapid-switch audio path: review_audio/Red To Blue/Wide Ratings/RapidSwitch.wav

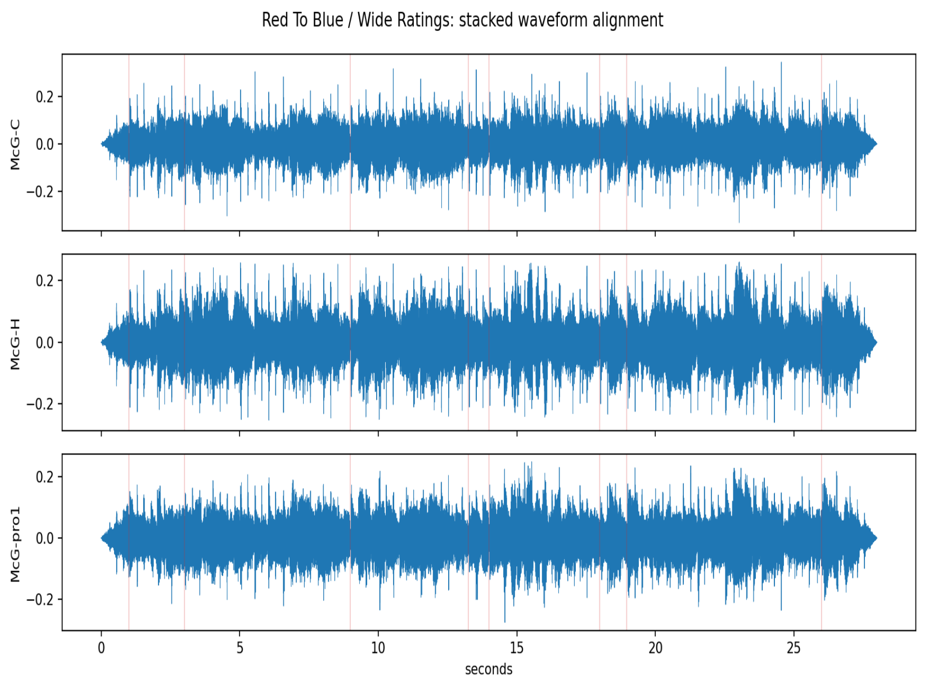
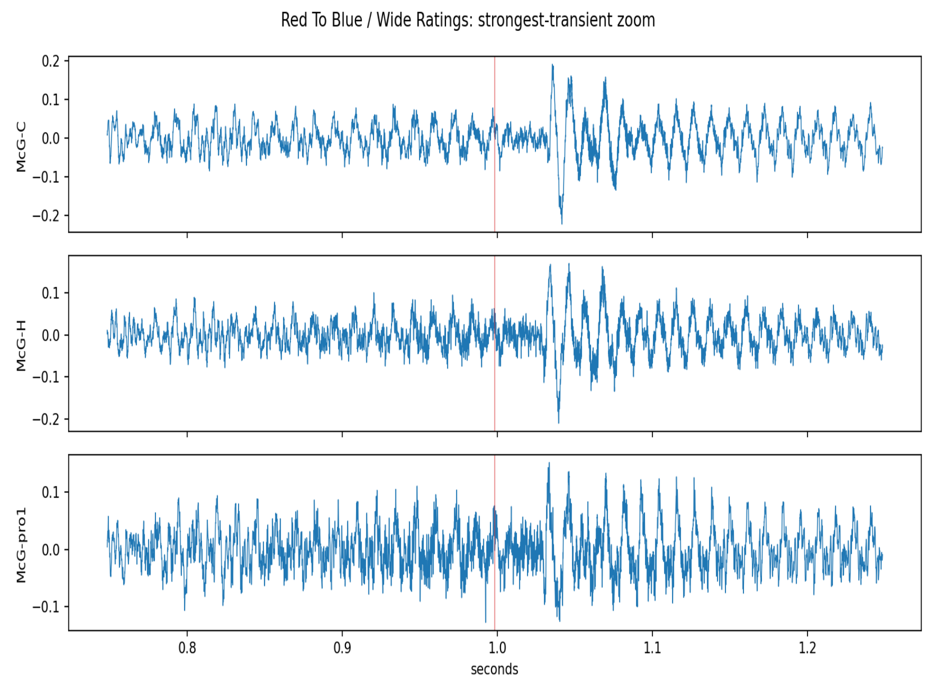
Pair	Lag samples	Lag ms	Correlation	Automatic result
McG-C-McG-H	0	0.000	0.833	PASS
McG-C-McG-pro1	0	0.000	0.710	PASS
McG-H-McG-pro1	0	0.000	0.734	PASS

Review-audio paths relative to 05_alignment_verification
review_audio/Red To Blue/Wide Ratings/McG-C_28sec.wav
review_audio/Red To Blue/Wide Ratings/McG-H_28sec.wav
review_audio/Red To Blue/Wide Ratings/McG-pro1_28sec.wav
review_audio/Red To Blue/Wide Ratings/RapidSwitch.wav

Manual-review prompts

- Do major transients align visually?
- Is there any apparent constant offset?
- Is there any timing drift across the 28-second excerpt?
- Does rapid switching reveal a perceptible timing jump?
- Is the issue alignment, or only a mix-production difference?
- Accept alignment?
- Reviewer comments.

Visual Evidence: Red To Blue - Wide Ratings

Full stacked-waveform figure	Zoomed strongest-transient figure
	
figures/Red To Blue/Wide Ratings/stacked_waveforms.png	figures/Red To Blue/Wide Ratings/strongest_transient_zoom.png

Manual-Review Checklist

Song	Condition	Mix names	Automatic result	Waveform checked	Rapid-switch checked	Audible issue	Accepted	Reviewer	Review date	Comments
In The Meantime	Similar Ratings	QUT-B | DU-H | DU-I	FAIL							
In The Meantime	Wide Ratings	DU-K | QUT-pro | DU-N	REVIEW							
Lead Me	Similar Ratings	DU-D | DU-E | QUT-F	FAIL							
Lead Me	Wide Ratings	PXL-L1 | PXL-L4 | McG-pro2	FAIL							
Pouring Room	Similar Ratings	McG-R | McG-T | McG-X	REVIEW							
Pouring Room	Wide Ratings	McG-pro1 | McG-U | McG-V	FAIL							
Red To Blue	Similar Ratings	McG-B | McG-G | McG-F	REVIEW							
Red To Blue	Wide Ratings	McG-C | McG-H | McG-pro1	PASS							

Final Status Summary

- Automatically passing triplets: Red To Blue / Wide Ratings.
- Triplets requiring REVIEW: In The Meantime / Wide Ratings, Pouring Room / Similar Ratings, Red To Blue / Similar Ratings.
- Triplets classified as FAIL/review-required: In The Meantime / Similar Ratings, Lead Me / Similar Ratings, Lead Me / Wide Ratings, Pouring Room / Wide Ratings.
- Maximum observed lag: 58.050 ms.
- Triplets with manual approval recorded: 0.
- Next action required: visual inspection of waveform figures and perceptual inspection of rapid-switch audio before final inclusion.
- The automatic classifications are screening results. Final inclusion requires visual inspection of the waveform figures and perceptual inspection of the rapid-switch audio.

Appendix

- Alignment thresholds used: PASS <=10 ms and correlation >=0.60; REVIEW <=30 ms and correlation >=0.35; otherwise FAIL.
- Lag fields report residual pairwise offsets in samples and milliseconds.
- Correlation fields report peak normalized cross-correlation over the Phase 2C search window.
- Canonical CSV files: tables/pairwise_alignment_verification.csv, tables/alignment_summary.csv, tables/manual_alignment_review.csv, tables/review_audio_manifest.csv.
- Figure files: figures///stacked_waveforms.png and figures///strongest_transient_zoom.png.
- Review-audio folders: review_audio/// with three individual 28-second WAVs plus RapidSwitch.wav.
- Review WAV hashes match Phase 2B source WAVs according to review_audio_manifest.csv.
- Protected upstream outputs remained unchanged during booklet generation.

Source	Path
pairwise_alignment_verification.csv	outputs/stimulus_selection/05_alignment_verification/tables/pairwise_alignment_verification.csv
alignment_summary.csv	outputs/stimulus_selection/05_alignment_verification/tables/alignment_summary.csv
manual_alignment_review.csv	outputs/stimulus_selection/05_alignment_verification/tables/manual_alignment_review.csv
review_audio_manifest.csv	outputs/stimulus_selection/05_alignment_verification/tables/review_audio_manifest.csv
recommended_triplets_for_review.csv	outputs/stimulus_selection/06_rating_stratification/tables/recommended_triplets_for_review.csv